Starting in equilibrium: energy and momentum fluctuations at a global temperature

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Abstract

The present thermal start (`thermal_init!`) draws the particle momenta at fixed particle energies, so the energy fluctuations are missing, and the stochastic step, which holds every particle energy fixed, never creates them. Here every particle energy and momentum is drawn from the equilibrium at one global β^μ: the maximum of the total entropy at fixed total energy and momentum, and the fluctuations about it. We define the global β^μ from the average totals, draw configurations whose totals fluctuate with the thermodynamic susceptibilities, optionally shift them to the exact targets so that a run needs no thermalization, and test the draws.

I. MAXIMUM ENTROPY AT FIXED TOTALS

At fixed positions each particle has a volume V_a, an energy e_a and a momentum $\boldsymbol{m}_a$, and its entropy is the function

$$S_a(e_a, \boldsymbol{m}_a) = V_a\, \sigma\big(e_a/V_a,\ \boldsymbol{m}_a/V_a\big), \tag{1}$$

with $\sigma(\varepsilon, \boldsymbol{M})$ the entropy density at energy density $\varepsilon = T^{00}$ and momentum density $\boldsymbol{M} = T^{0i}$ (`stochrep.tex`). The totals

$$E = \sum_a e_a, \qquad \boldsymbol{M}_{\rm tot} = \sum_a \boldsymbol{m}_a \tag{2}$$

are conserved. The most probable state maximizes $\sum_a S_a$ at fixed E and $\boldsymbol{M}_{\rm tot}$. With multipliers β^0 and $\boldsymbol{\beta}$, each particle maximizes the same function,

$$\Phi_a(e_a, \boldsymbol{m}_a) = S_a(e_a, \boldsymbol{m}_a) - \beta^0 e_a + \boldsymbol{\beta}\cdot\boldsymbol{m}_a, \tag{3}$$

so that at the maximum

$$\frac{\partial S_a}{\partial e_a} = \beta^0 = \frac{1}{T}, \qquad \frac{\partial S_a}{\partial \boldsymbol{m}_a} = -\boldsymbol{\beta} = -\frac{\boldsymbol{v}}{T}, \tag{4}$$

with $T = T_{\rm rel}/\gamma$ the bare temperature and $\boldsymbol{v}$ the fluid velocity, common to all particles. In covariant form, with $\beta_\mu = u_\mu/T_{\rm rel}$ and the metric $(-,+,+)$, $\Phi_a = S_a + \beta_\mu T_a^{0\mu}$, since $\beta_0 = -\gamma/T_{\rm rel} = -\beta^0$ and $\beta_i = u^i/T_{\rm rel} = \beta^i$. The particle quantity in eq. (3) is the entropy function S_a of the particle's energy and momentum, not the entropy s_a itself; $s_a = S_a(e_a, \boldsymbol{m}_a)$ once e_a and $\boldsymbol{m}_a$ are drawn.

II. FLUCTUATIONS AND THE GLOBAL β^μ

a. One particle. In equilibrium the probability of a configuration is $e^{\sum_a S_a}$ at the fixed totals. For a system of many particles each particle is then distributed independently as

$$p_a(e, \boldsymbol{m}) \propto e^{\Phi_a(e,\boldsymbol{m})} = e^{S_a(e,\boldsymbol{m}) - \beta^0 e + \boldsymbol{\beta}\cdot\boldsymbol{m}}. \tag{5}$$

To quadratic order about the maximum, $x = (\delta e, \delta\boldsymbol{m})$ is Gaussian with covariance

$$C_a = \big(-\partial^2 S_a\big)^{-1}, \tag{6}$$

the particle's susceptibility. At rest, with $c_v = d\varepsilon/dT$ and $w = e + p$,

$$\langle \delta e^2\rangle = V_a T^2 c_v, \qquad \langle \delta m^i \delta m^j\rangle = V_a\, wT\, \delta^{ij}. \tag{7}$$

For the conformal equation of state $V_a T^2 c_v = 4 V_a \varepsilon T$.

b. The average is not the maximum. The distribution eq. (5) is skewed: the mean of e lies above its most probable value by 0.16 standard deviations at $s_a = 720$ and by 0.58 at $s_a = 54$, and at $s_a = 54$ the momentum variance is 16% above eq. (6). We therefore define the global β^μ by the *average* totals,

$$\sum_a \langle e_a \rangle_\beta = E, \qquad \sum_a \langle \boldsymbol{m}_a \rangle_\beta = \boldsymbol{M}_{\rm tot}. \tag{8}$$

With $\theta = (-\beta^0, \boldsymbol{\beta})$ conjugate to $x = (e, \boldsymbol{m})$, eq. (5) is an exponential family, so $\partial\langle x\rangle/\partial\theta = \mathrm{Cov}(x)$ exactly, and eq. (8) is solved by Newton's method,

$$\theta \to \theta + \Big[\sum_a \mathrm{Cov}_a(x)\Big]^{-1} \Big[(E, \boldsymbol{M}_{\rm tot}) - \sum_a \langle x_a \rangle\Big], \tag{9}$$

starting from the most probable state at the mean densities. The resulting T is 0.5% below the temperature of the most probable state at $s_a = 541$ and 5% below at $s_a = 54$.

c. Fluctuating or exact totals. The particles are drawn independently from eq. (5), the canonical ensemble; the totals then fluctuate about $(E, \boldsymbol{M}_{\rm tot})$ with the global susceptibility

$$\mathrm{Cov}\big(E, \boldsymbol{M}_{\rm tot}\big) = \sum_a \mathrm{Cov}_a(x) \;\simeq\; V_{\rm tot}\, \chi, \tag{10}$$

$V_{\rm tot} = \sum_a V_a$ and χ the susceptibility of $T^{0\mu}$ per volume. To start an SPH run at exactly the target E and $\boldsymbol{M}_{\rm tot}$, the difference between the drawn totals and the targets is then removed uniformly,

$$x_a \to x_a - \frac{V_a}{V_{\rm tot}} \Big[\sum_b x_b - (E, \boldsymbol{M}_{\rm tot})\Big], \tag{11}$$

a uniform shift of the energy and momentum densities (an equal share per particle on an equal-volume lattice). This changes each particle's distribution only at order $1/N$, so the run can start without a thermalization step.

III. IMPLEMENTATION

In `src/measure.jl`:

- `particle_moments`: the exact $\langle x\rangle$ and $\mathrm{Cov}(x)$ of eq. (5) by quadrature on a 41^3 grid around the maximum (-6 to $+10$ standard deviations in e, ± 9 in $\boldsymbol{m}$); particles of equal volume share them.
- `beta_from_averages`: the Newton solution of eq. (8).
- `draw_particle`: one exact draw from eq. (5), an independence Metropolis chain of 30 steps whose proposal is the Gaussian of eq. (6) with twice its covariance (wide enough to cover the skewed tail); the acceptance is 0.59.
- `equilibrium_init!(ps, params, cl, eos; exact_totals = false)`: from the totals of the current state (or given E, $\boldsymbol{M}_{\rm tot}$), solves for β^μ, draws $(e_a, \boldsymbol{m}_a)$ at the current volumes $V_a = s_a/S_a$, sets $s_a = S_a(e_a, \boldsymbol{m}_a)$ and re-smooths. With `exact_totals = true` the totals are then shifted to the targets as above.

IV. TESTS

`test/testeqstart.jl`; conformal equation of state, $h/a = 2$, box 0.6 ($N = 672$), $V_a = 5.4 \times 10^{-4}$.

a. One particle. The draws of `draw_particle` (4×10^4) agree with direct quadrature of eq. (5) in mean, variances and the e–m_x covariance, at rest and at $v = 0.5$, at $s_a \simeq 700$ and at $s_a \simeq 50$, within the statistical errors; the maximum satisfies eq. (4) to 10^{-6}.

b. The lattice. With `exact_totals = true` the totals are exact to 10^{-12}, the averages at the solved β^μ reproduce them, and the spread of $(e_a, \boldsymbol{m}_a)$ over the particles equals the single-particle covariance (at rest and at $v = 0.32$).

c. Global susceptibility. 3000 canonical configurations of the lattice:

	$\langle E\rangle/E - 1$	$\frac{\mathrm{Var}\,E}{\sum_a C_a^{ee}}$	$\frac{\mathrm{Var}\,M_x}{\sum_a C_a^{xx}}$	$\mathrm{corr}(E, M_x)$	predicted
$s_a = 541$, rest	$(-1.3 \pm 0.7) \times 10^{-4}$	0.988	0.968	-0.003	0
$s_a = 541$, $v = 0.32$	$(-1.4 \pm 0.7) \times 10^{-4}$	0.996	0.975	0.724	0.727
$s_a = 54$, rest	$(-2.0 \pm 2.0) \times 10^{-4}$	0.967	0.991	-0.008	0

(statistical error of the variance ratios ± 0.026). At rest and $s_a = 541$, $\mathrm{Var}\,E = 0.993\,V_{\mathrm{tot}}T^2c_v$ and $\mathrm{Var}\,M_x = 0.983\,V_{\mathrm{tot}}wT$; at $s_a = 54$, $\mathrm{Var}\,M_x = 1.17\,V_{\mathrm{tot}}wT$, the non-Gaussian excess, which eq. (10) with the exact Cov_a includes. The two means at $s_a = 541$ lie 2σ low with the same sign, a possible bias of order 10^{-4} to be checked with more configurations.

V. THE SPH REPRESENTATION AFTER THE DRAW

The draw holds the volumes V_a fixed. After re-smoothing, the code's volumes are $V'_a = s_a/S'_a$, which follow the drawn entropy: a particle with more entropy has more volume at nearly the same smoothed density. Its energy recomputed from the SPH state, $e'_a = s_a\,\mathcal{E}(S'_a, \dots)$, then changes by

$$\delta e'_a = \frac{\varepsilon}{S}\,\delta s_a = \tfrac{3}{4}\,T\,\delta s_a \qquad \text{instead of} \qquad \delta e_a = T\,\delta s_a \tag{12}$$

at fixed momentum (conformal, $\varepsilon = \frac{3}{4}TS$), so $\mathrm{Var}\,e'/\mathrm{Var}\,e \simeq 9/16$. Measured: 0.62, with $\mathrm{corr}(e', e) = 0.997$ and $\mathrm{rms}(V'/V - 1) = 0.07$.

In a coupled run (leapfrog and stochastic step, $S = 10^6$) the variance of the recomputed particle energies, in units of the drawn one, stays at 0.57–0.61 over $t = 3$ from this start, and at 0–0.004 from `thermal_init!`: on this time scale the dynamics neither create nor remove these fluctuations, so they must be put in at the start. Which energy variance is the equilibrium one for the particle representation, $\mathrm{Var}\,e$ or $\frac{9}{16}\mathrm{Var}\,e$, is open.